Computer Vision Project

Efhamni

Tuwaiq Academy – Building and Developing AI Models Bootcamp

Ruwaa Surrati
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Team Members:

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Problem Statement

Communication isn't always easy. Not everyone knows sign language, and even among those who do, different countries often use different versions. At the same time, not everyone can hear, and not everyone can read subtitles comfortably. This creates real barriers for people who rely on sign language in their daily lives.

Our Solution

That's where Efhamni comes in. Our project takes Arabic Sign Language gestures, translates them into readable Arabic subtitles, and then uses text-to-speech so the subtitles are spoken out loud. In short, it helps bridge the communication gap by turning signs into both text and sound.

Dataset

We're using the Saudi Sign Language Library, provided by the Saudi Association for Hearing Impairment. This gives us access to videos of words, letters, numbers, and phrases in Saudi Sign Language, which will serve as the foundation for training our models.

Project Workflow

1. Data Preparation & Preprocessing

- **Hisham** used AIOHTTP Python scraper to collect the videos (.mp4) of the speakers gesturing the Arabic Sign Language words, letters, numbers, and phrases.
- **Everyone** will manually validate the dataset to make sure all videos are extracted correctly:
    - **Abdullah**: Letters
    - **Hisham**: Numbers
    - **Rayid**: Words
    - **Ruwaa**: Sentences

- **Abdullah** will handle keypoint extraction (hands, hand positions, and gesture movements) using MediaPipe.
- **Rayid** will apply normalization and labeling to prepare the keypoints for training.

2. Sign Recognition Model

- **Ruwaa** will train a CNN+LSTM (or Transformer) to recognize signs from sequences of frames.
- Keypoints will be fed into the sequence model (LSTM/GRU/Transformer) to learn the structure of each sign.
- The model will output recognized words, which will later be ordered into meaningful sentences.

4. Subtitle Generation & Text-to-Speech

- *…Work in progress…*

5. User Interface

- *…Work in progress…*

Problems We Faced	How We Overcame Them
No existing ready-to-use dataset	Collected videos by scraping the Saudi Sign Language Library
Mixed file formats (.gif, .png) that include irrelevant data like the headers and website logo	Filtered the dataset and kept only valid and relevant .mp4 files
Only one instance per sign	Will apply video augmentation techniques to expand the dataset and add variation
Data not sequenced (signs may form reversed sentences)	*…Work in progress…*